

1. The hyperdegree distribution of each multilayer hypergraph conforms to the power law distribution.

The hyperdegree distribution of each multilayer hypergraph is shown in Figure 1.

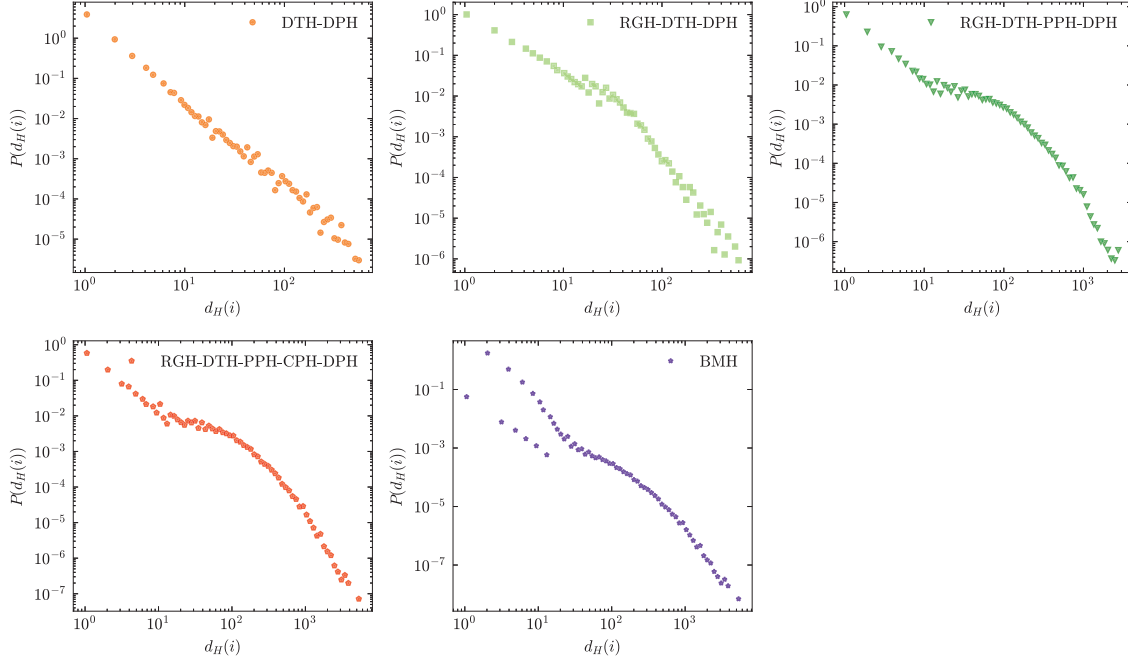


Fig. 1. Scale-free properties of various multilayer hypergraph

2. Distribution of the shortest path length in each multilayer hypergraph

The distribution of the shortest path lengths of all drug-disease pairs is shown in Figure 2. As can be seen from Figure 2, except for the DTH-DPH two-layer hypernetwork with a path length of 3, the path distributions of other hierarchical hypernetworks all follow a Poisson distribution, and the shortest paths are most distributed on the paths with 3 nodes in the middle (shortest path length = 5).

3. Performance comparison between hyperiDPath and iDPath

Both hyperiDPath and iDPath (2) were implemented using Python 3.9, PyTorch 1.12.0, xgi 0.8.2 (1), scikit-learn 1.4.0, and RTX 4090. All models in the experiment were trained by the Adam optimizer, with the ratio of training set, validation set, and test set being 8:1:1. The hyperparameters were determined by optimizing the AUROC on the validation set. The hyperparameters of hyperiDPath and iDPath were set as follows: the batch size, learning rate, the number of shortest paths L for a drug-disease pair, and the fixed length l_{max} of all shortest paths were set to 512, 0.001, 256, and 7, respectively; the dropout was set to 0.5, the embedding dimension and the size of the hyperGCN layer were 64 and 64-128-64, and the size of the LSTM layer was 64-64. The experimental results of each dataset are detailed in Table 1.

4. External validation: performance comparison between hyperiDPath and iDPath

The purpose of external validation is to test the performance of hyperiDPath and iDPath on other datasets. We use the multi-layer biological network dataset of iDPath (2) for performance comparison. The results of comparing the indicators of hyperiDPath and iDPath on various datasets are shown in Table 2.

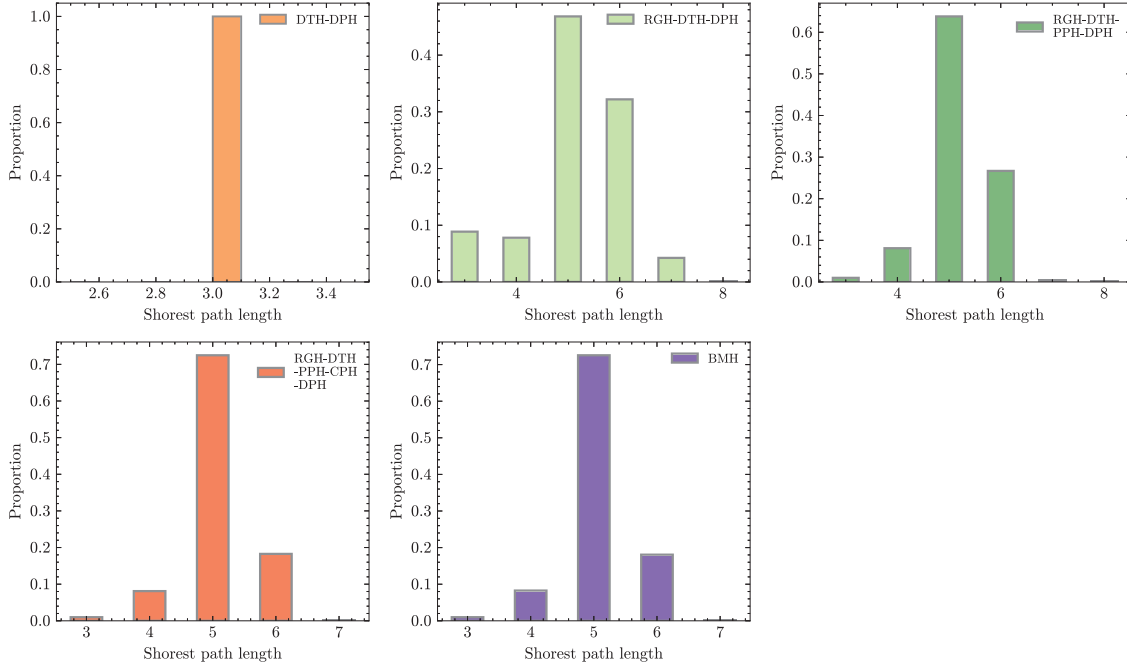


Fig. 2. Poisson distribution of the shortest path length in each layer of the hypergraph

Table 1. Comparison of performance indicators of hyperiDPath and iDPath models. Bold numbers indicate the best performance

Datasets	Models	Accuracy	Recall	AUROC	AUPRC	F1-Score
BMH	hyperiDPath	0.8340	0.8614	0.9043	0.8851	0.8365
	iDPath	0.8265	0.8483	0.8990	0.8849	0.8279
BioKG	hyperiDPath	0.7754	0.7909	0.8477	0.8443	0.7834
	iDPath	0.7700	0.8137	0.8429	0.8384	0.7840
Openbiolink	hyperiDPath	0.7291	0.7656	0.7903	0.8008	0.7410
	iDPath	0.6706	0.6799	0.7518	0.7853	0.6876

Table 2. Comparison of performance metrics of hyperiDPath and iDPath models on iDPath literature data. Bold numbers indicate the best performance

Models	Accuracy	Recall	AUROC	AUPRC	F1-Score
hyperiDPath	0.9263	0.9262	0.9720	0.9743	0.9280
iDPath	0.9191	0.9230	0.9685	0.9708	0.9213

5. T-test of hyperiDPath and iDPath

In order to evaluate whether the performance difference between hyperiDPath and iDPath is statistically significant, we conducted 20 independent experiments on the OpenBiolink dataset for the two methods and collected relevant performance indicators (such as AUC, AUPR). Subsequently, the independent samples t-test was used to perform statistical analysis on the two sets of experimental results to test whether the difference between the two was significant. The experimental results are shown in Figure 3. Both methods showed extremely significant statistical differences in both AUC and AUPRC (p values were much less than 0.001), which fully demonstrated that hyperiDPath was significantly better than iDPath in performance.

6. Discussion on different path generation methods

In the hypergraph representation learning task, the path generation method has an important impact on the performance of the model. We systematically compared the shortest path of the hypergraph s-walk(3), the shortest path of the hypergraph to bipartite graph, and the graph shortest path on the performance of hyperiDPath in the BMH dataset. The experimental results show that, except for the graph shortest path, the paths generated by the other methods failed to achieve results comparable to the shortest

Performance Comparison of hyperiDPath and iDPath

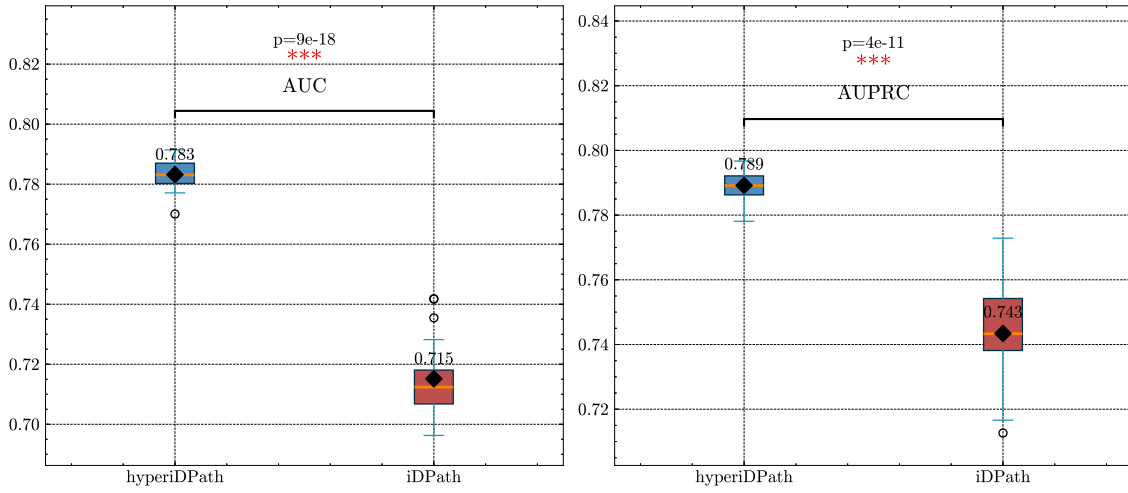


Fig. 3. hyperiDPath vs iDPath is statistically significant

path. This shows that the graph shortest path has obvious advantages in capturing the structural relationship between nodes and improving model performance. The specific results are shown in Figure 4 below.

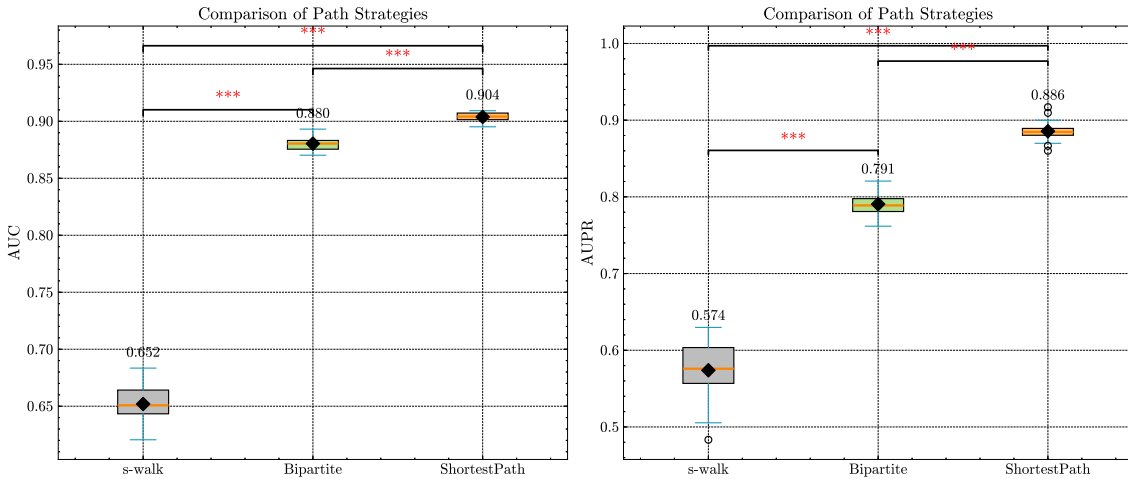


Fig. 4. Effects of different path generation methods on the results

As shown in Figure 4, the graph shortest path is significantly better than the hypergraph to bipartite graph shortest path and hypergraph s-walk methods in both AUC and AUPR evaluation indicators. Specifically, the s-walk path frequently passes through virtual hyperedges such as PPI and CCI in multi-layer hypergraphs, which easily introduces a lot of noise and false connections, resulting in overfitting of the model. Its prediction performance is significantly lower than other methods, and the results fluctuate greatly. Although the bipartite graph path is essentially similar to the shortest path of the ordinary graph, it can improve some performance by skipping virtual hyperedges, but its computational and storage overhead is significantly increased, and in some experiments it is still slightly inferior to the shortest path calculated directly on the graph. From a statistical analysis, the differences in AUC and AUPR among the three methods are all extremely significant ($p < 0.001$), further verifying the key role of path generation strategy in the learning effect of hypergraph representation. The overall results show that the reasonable selection of path generation methods, especially the use of graph shortest paths, can more effectively capture the essential structural relationship between nodes and improve the generalization and prediction ability of the model for complex biological networks.

7. Discussion on biological functions

To further enhance the representational capabilities of the model, we introduced a biological function layer based on the original BMH dataset and constructed two additional hypergraphs: the biological function-protein hypergraph (BPH) with 7993 nodes and 6290 hyperedges, and the biological function-biological function hypergraph (BBH) with 9798 nodes and 22545 hyperedges (4). This design aims to improve the contextual expression capabilities of protein nodes by introducing richer semantic relationships. The experimental results are shown in Figure 5.

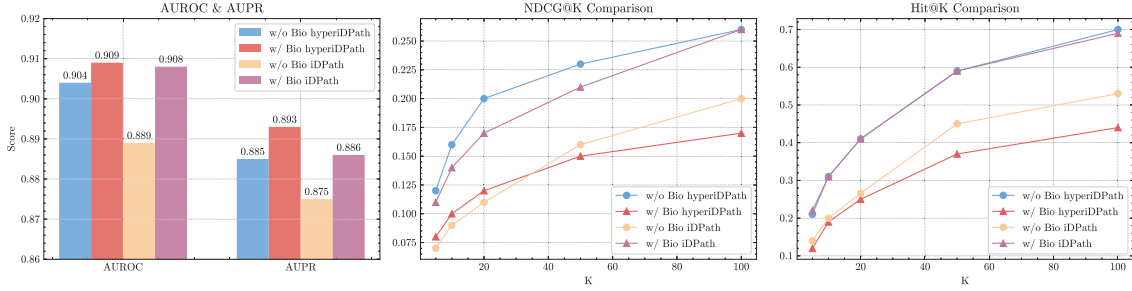


Fig. 5. Effect of the introduction of biological functional layer on model classification and recommendation performance

Figure 5 compares the performance of the hyperiDPath and iDPath models before and after the introduction of the biological functional layer on the BMH dataset. The experimental results show that there are significant differences in the responses of the two models to the biological functional layer under different tasks and evaluation indicators.

In terms of classification tasks (AUROC and AUPR), the hyperiDPath model always outperforms iDPath regardless of whether biological function-related data is introduced, and both have a slight improvement after adding the biological function layer. Specifically, the AUROC and AUPR of hyperiDPath without the introduction of the biological function layer were 0.904 and 0.885, respectively, both higher than iDPath's 0.889 and 0.875; after the introduction of the biological function layer, these two indicators of hyperiDPath were further improved, and continued to maintain its lead over iDPath. This shows that hyperiDPath can more effectively mine the global structure and semantic information in the protein network, and show stronger discrimination ability in the protein-disease association classification task.

In terms of recommendation system related indicators (NDCG@K, Hit@K), the two models show completely different trends in the introduction of the biological function layer. After integrating the biological function information, the recommendation performance of iDPath is significantly improved, indicating that its structure is more suitable for utilizing the high-order semantic information brought by the biological function layer. In contrast, after the introduction of the biological function layer, the NDCG@K and Hit@K indicators of hyperiDPath have both decreased to a certain extent, reflecting that it is more sensitive to redundant or irrelevant information when processing complex hypergraph structures.

We speculate that the main reason for this phenomenon is that there is a lack of direct correlation between the newly added biological function nodes and the disease, which makes the calculation of the shortest path more likely to fall into the local optimum, thereby weakening the global transmission efficiency of structural information. In addition, the number of edges in the biological function layer in the multi-layer hypergraph is significantly less than that of the core layers such as PPH and CCH, which further increases the structural noise and has a negative impact on the model's discriminative ability.

Therefore, the advantage of hyperiDPath is mainly reflected in its ability to model the global structure of protein-disease relationships in classification tasks, but it is more sensitive to information quality when introducing biological function information. Therefore, optimizing structural design and information screening strategies is crucial to maximize its potential. After integrating biological function layer information, iDPath's recommendation performance improved more significantly, showing that its structure has better adaptability to multi-source high-order semantic information. In the future, it is recommended to combine the advantages of the two types of models and further optimize the information introduction method to improve the overall performance under different tasks.

8. Different batch sizes, lengths and numbers of shortest paths are related to the performance of hyperiDPath

The hyperiDPath classification task under different batches and different shortest path numbers (l_{max}) on the BMH dataset was tested, as shown in Figure 6(a-e); under the optimal parameters, the effect of different shortest path lengths on the hyperiDPath was tested, as shown in Figure 6(f). According to Figure 6(a-e), under different hyperparameters, except for the large fluctuation of the recall curve, the other indicators of hyperiDPath are higher than those of iDPath. When the batch size is 64, the AUROC indicator is high, but the time consumption is long, while when the batch size is 512 and the shortest path number is 256, the performance is good and the time consumption is short. In Figure 6(f), the experiment with the shortest path length $L_{length} = 6$ cannot be calculated due to out of memory. When $L_{length} = 7$, the performance is best. This is because the maximum length of the shortest path is set to 7, which can cover almost all drug-disease pairs.

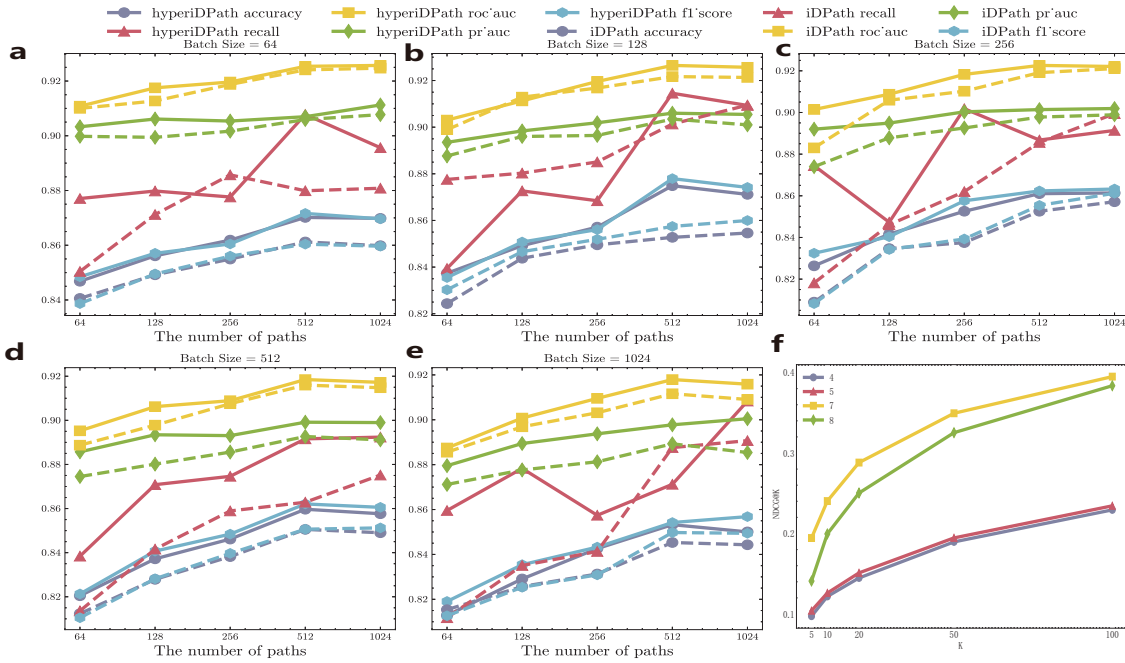


Fig. 6. Effects of batch size, number of paths, and maximum path length on the performance of hyperiDPath. a-e. Performance comparison of hyperiDPath with different batch sizes and different numbers of shortest paths. f. Statistical analysis of the shortest path number (l_{max}) of hyperiDPath when other hyperparameters are fixed.

10. hyperiDPath time complexity analysis

In order to systematically evaluate the scalability and practical application value of the model in large-scale biological knowledge graphs, we further compared the actual running time of hyperiDPath and iDPath on three real biological knowledge graph datasets: OpenBioLink, BioKG, and BMH. The experimental results are shown in Figure 7.

Although hyperiDPath improves the representation of higher-order biological interactions, it inevitably introduces additional computational overhead compared with the original iDPath framework. This increased cost mainly arises from three components: dual hypergraph convolution, LSTM-based sequential path aggregation, and multi-granularity attention learning. As shown in Figure 7, hyperiDPath requires longer runtime than iDPath across all three datasets. On OpenBioLink and BioKG, the runtime increased moderately from 775 s to 896 s and from 2748 s to 3201 s, respectively. A more substantial increase was observed on the BMH dataset, where the runtime increased from 2978 s to 13141 s. This result indicates that the computational burden of hyperiDPath becomes more pronounced when applied to dense multilayer hypergraphs with richer higher-order structures.

The additional runtime is primarily attributable to the construction and propagation of hyperedge-level representations. Unlike pairwise graph models, hypergraph learning needs to update both entity-level and hyperedge-level embeddings through the incidence

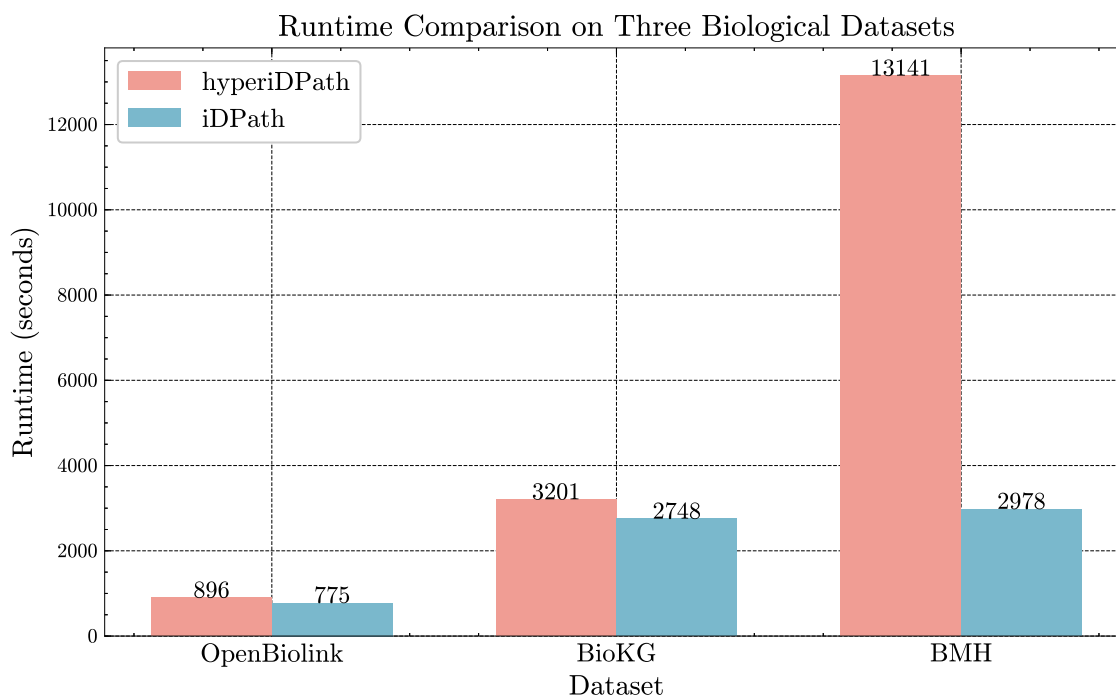


Fig. 7. Comparison of the running time of hyperiDPath and iDPath on three biological knowledge graphs of different scales. As the amount of data increases, the computational cost of hyperiDPath increases significantly, especially on the BMH dataset, where the running time is more than four times that of iDPath, reflecting its computational bottleneck in large-scale graphs

structure. Moreover, mechanistic path reasoning requires sequential aggregation over candidate MODA paths, and the multi-level attention modules further increase the cost of representation weighting and path scoring. Therefore, the improved modeling capacity of hyperiDPath is achieved at the expense of higher time and memory consumption.

Despite this limitation, several strategies can be adopted to improve scalability. First, hypergraph construction and MODA path extraction can be performed as offline preprocessing steps, thereby avoiding repeated real-time computation during model training. Second, simplified hypergraph convolution operators inspired by decoupled graph propagation, such as SGCN-style propagation, may reduce the computational complexity of message passing while preserving higher-order structural information. Third, sub-hypergraph sampling or mini-batch hyperedge sampling can be introduced to restrict each training iteration to a local region of the multilayer hypergraph, reducing memory consumption and improving training efficiency. Finally, parallel computation of candidate paths and attention scores can further accelerate large-scale applications.

Therefore, while hyperiDPath is computationally more expensive than iDPath, the increased cost reflects the additional modeling of higher-order biomolecular structures. Future work will focus on developing more efficient hypergraph propagation and sampling strategies to improve scalability for genome-scale biomedical networks.

References

- Landry, Nicholas W and Lucas, Maxime and Iacopini, Iacopo and Petri, Giovanni and Schwarze, Alice and Patania, Alice and Torres, Leo. XGI: A Python package for higher-order interaction networks. *Journal of Open Source Software*, 8(85):5162, 2023.
- Yang, Jiannan and Li, Zhen and Wu, William Ka Kei and Yu, Shi and Xu, Zhongzhi and Chu, Qian and Zhang, Qingpeng. Deep learning identifies explainable reasoning paths of mechanism of action for drug repurposing from multilayer biological network. *Briefings in Bioinformatics*, 23(6):bbac469, 2022.
- Aksoy, Sinan G and Joslyn, Cliff and Marrero, Carlos Ortiz and Praggastis, Brenda and Purvine, Emilie. Hypernetwork science via high-order hypergraph walks. *EPJ Data Science*, 9(1):16, 2020.
- Ruiz, Camilo and Zitnik, Marinka and Leskovec, Jure. Identification of disease treatment mechanisms through the multiscale interactome. *Nature Communications*, 12(1):1796, 2021.